

KUNAL M SHENDE

Embedded Systems Engineer | IoT Security Researcher

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PROFESSIONAL SUMMARY

Computer Science undergraduate with 2 years of experience in embedded systems, Internet of Things (IoT) hardware security, and web development. Proficient in vulnerability assessment, penetration testing, and firmware engineering. National competition award winner and Indian Design Patent holder.

TECHNICAL SKILLS

Programming Languages: C, Embedded C, C++, Python, TypeScript, Bash

IoT and Embedded Hardware: ESP32, ESP32-S3, Arduino, Raspberry Pi, FreeRTOS, SPI, I2C, UART

Web Development: HTML5, CSS3, Bootstrap, Python Flask, WebSockets, Firebase

Databases: MySQL, Supabase, Firebase Firestore

Cybersecurity Tools: Burp Suite, Nmap, Wireshark, Metasploit, OWASP Top 10

DevOps and Cloud: Git, GitHub, Docker, Vercel, Google Cloud Platform, Visual Studio Code, Postman

WORK EXPERIENCE

Independent Security Researcher | *Freelance* | January 2024 - Present

- Architected 3 secure IoT prototypes on ESP32-S3 and ATmega328P, reducing firmware attack surfaces by 40 percent.
- Discovered 15 critical vulnerabilities in 10 controlled environments using OWASP and NIST assessment frameworks.
- Reduced network anomalies by 25 percent via packet-level traffic analysis and proactive hardening of 5 IoT deployments.

Web Exploit Hunting Intern | *EduSkills and AICTE* | June 2026 - August 2026

- Detected 12 security flaws including SQL injection and Cross-Site Scripting in 8 web applications during penetration testing.
- Compiled detailed vulnerability reports with severity ratings, accelerating remediation timelines by 30 percent.

Cybersecurity Intern | *Palo Alto Networks and EduSkills* | January 2026 - March 2026

- Analyzed 500 packet captures and firewall rule sets, strengthening endpoint protection for 3 cloud environments.
- Earned 4 industry certifications from Palo Alto Networks in Security Operations, Network Security, and Cloud Security.

Java Full Stack Developer Virtual Intern | *EduSkills and AICTE* | October 2025 - December 2025

- Completed a 12-week program covering full-stack application development, database integration, and backend service design.

Cloud Virtual Intern | *AWS Academy* | April 2025 - June 2025

- Studied 10 core Amazon Web Services including EC2, S3, VPC, and IAM over an 8-week hands-on lab curriculum.

Android Developer Virtual Intern | *Google for Developers* | January 2025 - March 2025

- Explored 8 Android development modules on Material Design, activity lifecycle management, and local data persistence.

PROJECTS

ZeroTrace - Cybersecurity Audit Device | *ESP32-S3, FreeRTOS, C++* | [\[GitHub Repo\]](#) | 2026

- Engineered an embedded audit device integrating 4 security modules for radio frequency scanning, Wi-Fi auditing, and USB injection.
- Optimized display rendering to 60 frames per second by distributing FreeRTOS tasks on dual processor cores.
- Secured 2nd Prize at TECHTANTRA 2K26 national competition among 50 participating teams.

Sentinel - HID Attack Detection Framework | *Python, Windows API* | 2026

- Developed a real-time USB threat detection system that neutralizes rogue Human Interface Devices within 10 milliseconds.
- Validated the framework against 100 percent of simulated USB attack vectors using automated forensic snapshots.

MicroFlasher - Web Serial Flashing Platform | *TypeScript, Web Serial API* | 2026

- Deployed a browser-based firmware flasher that eliminated native toolchain setup and reduced onboarding time by 70 percent.

Autonomous Line-Follower Robot | ATmega328P, L298N, IR Sensors | [\[GitHub Repo\]](#) | 2025

- Constructed a differential-drive robot with Pulse Width Modulation calibration at 7.8 kilohertz, reducing motor noise by 90 percent.
- Calibrated a proportional control algorithm that achieved 95 percent path accuracy on sharp 90-degree turns.

ACHIEVEMENTS

- Won 2nd Prize at TECHTANTRA 2K26 National Level Technical Competition among 50 teams for the ZeroTrace project.
- Filed Indian Design Patent (Application Number 499678-001, CBR Number 209757) in April 2026 for ZeroTrace hardware.

EDUCATION

Bachelor of Technology in Computer Science and Engineering | S.B. Jain Institute of Technology, Nagpur | August 2023 - June 2027
Specialization: IoT Architecture, Cybersecurity, Embedded Systems

Higher Secondary Certificate, Science | Municipal Junior College, Kalmeshwar | June 2021 - March 2023

Secondary School Certificate, Distinction | Municipal Junior College, Kalmeshwar | June 2019 - March 2021

CERTIFICATIONS

- Cyber Security Workshop - Crow Security and IIT Bombay, 2025-2026
- Ethical Hacking and Penetration Testing - NASSCOM FutureSkills Prime, 2025-2026
- Security Operations Fundamentals - Palo Alto Networks, March 2026
- Network Security Fundamentals - Palo Alto Networks, March 2026
- Cybersecurity Fundamentals - Palo Alto Networks, March 2026
- Cloud Security Fundamentals - Palo Alto Networks, March 2026
- Python Basic and Problem Solving - HackerRank, 2025
- AWS Cloud Computing Fundamentals - AWS Academy, 2025
- German Language Proficiency Level 1 - FLOA, December 2025

LANGUAGES

English (Professional) | Hindi (Native) | Marathi (Native) | German (Beginner)